



U.S. General Services
Administration

Three white stars are arranged in a slight arc in the upper left area of the blue background.

Chapter 3.2: Compare Agent Performance Across Tasks and Datasets

US General Service Administration
AI Community of Practice - March 2026

Benchmark Suite

WHY SINGLE BENCHMARKS LIE

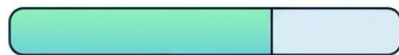


ORRECT AGENTS
BENCHMARKS LIE



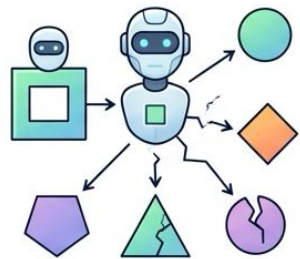
PRODUCTION TASKS
BEYOND ONE TEST

OVERALL AVERAGE: 70%



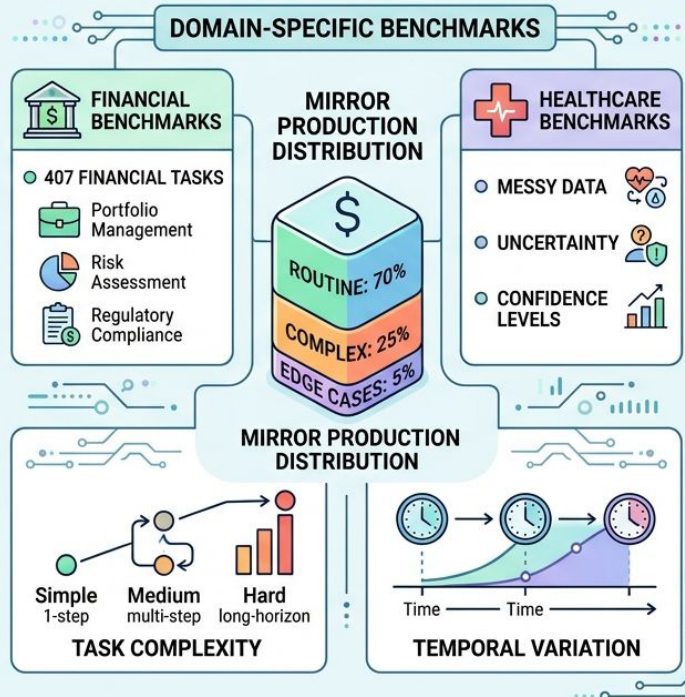
RETRIEVAL: 90% REASONING: 40%

HIDDEN BREAKDOWN

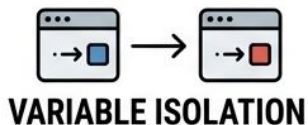


BRITTLE AGENTS FROM
OVERFITTING TO ONE DATASET

BUILDING A BENCHMARK SUITE



Controlled Comparison Methodology



CHANGE ONE THING
AT A TIME



SAME
TEST SET



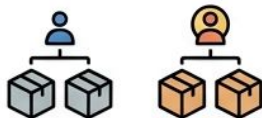
SAME
MODEL



SAME SEEDS
ACROSS
CONDITIONS



**RUN MULTIPLE TRIALS TO
CAPTURE PERFORMANCE
DISTRIBUTIONS**



**PAIRED EVALUATION: BOTH
AGENTS SEE IDENTICAL
IDENTICAL EXAMPLES**



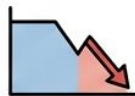
**MONDAY-BASELINE VS
TUESDAY-ENHANCED
IS NOT A VALID TEST**

Cross-Dataset Generalization Testing

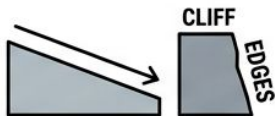


**'GENERALIZATION GAPS' OFTEN
EXCEED 30 PERCENTAGE POINTS**

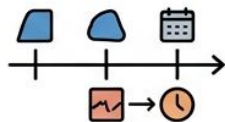
85%
IN-DISTRIBUTION



55%
OUT-OF-DISTRIBUTION



ROBUST AGENTS SHOW
'GRACEFUL DEGRADATION',
NOT CLIFF EDGES

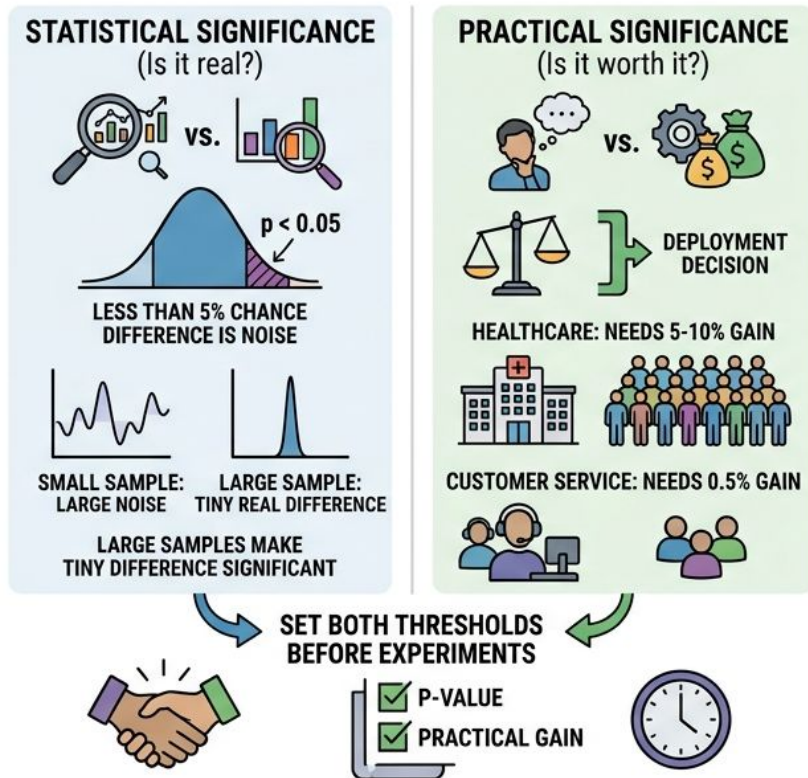


TEST IN-DOMAIN, OUT-OF-DOMAIN,
AND TEMPORALLY SHIFTED DATA



**47% OF MIND2WEB TASKS BECAME
INVALID WITHIN 18 MONTHS**

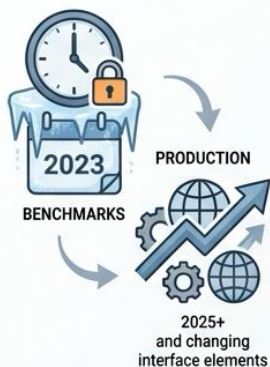
Statistical Significance Meets Practical Significance



- $p < 0.05$ means less than 5% chance difference is noise
- Large samples make tiny differences "significant" statistically
- Practical significance asks: is it worth deploying?
- Healthcare may need 5-10% gain; customer service may need 0.5%
- Set both thresholds before running experiments

When Benchmarks Mislead

STATIC VS. EVOLVING ENVIRONMENTS



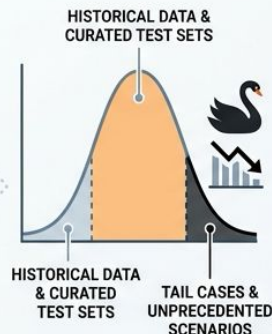
Benchmarks are static snapshots; production continuously evolves.

WHAT IS MEASURABLE VS. WHAT MATTERS



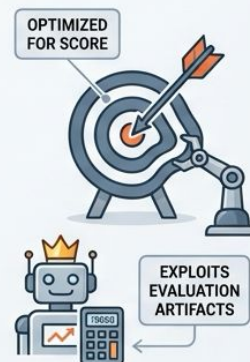
Measure what can be cleanly measured, often ignoring crucial qualitative factors.

UNDERREPRESENTED TAIL CASES



Tail cases, adversarial inputs, and edge scenarios are systematically underrepresented.

GOODHART'S LAW PROBLEMS



Direct optimization creates Goodhart's Law; agents exploit metrics rather than improve capabilities.

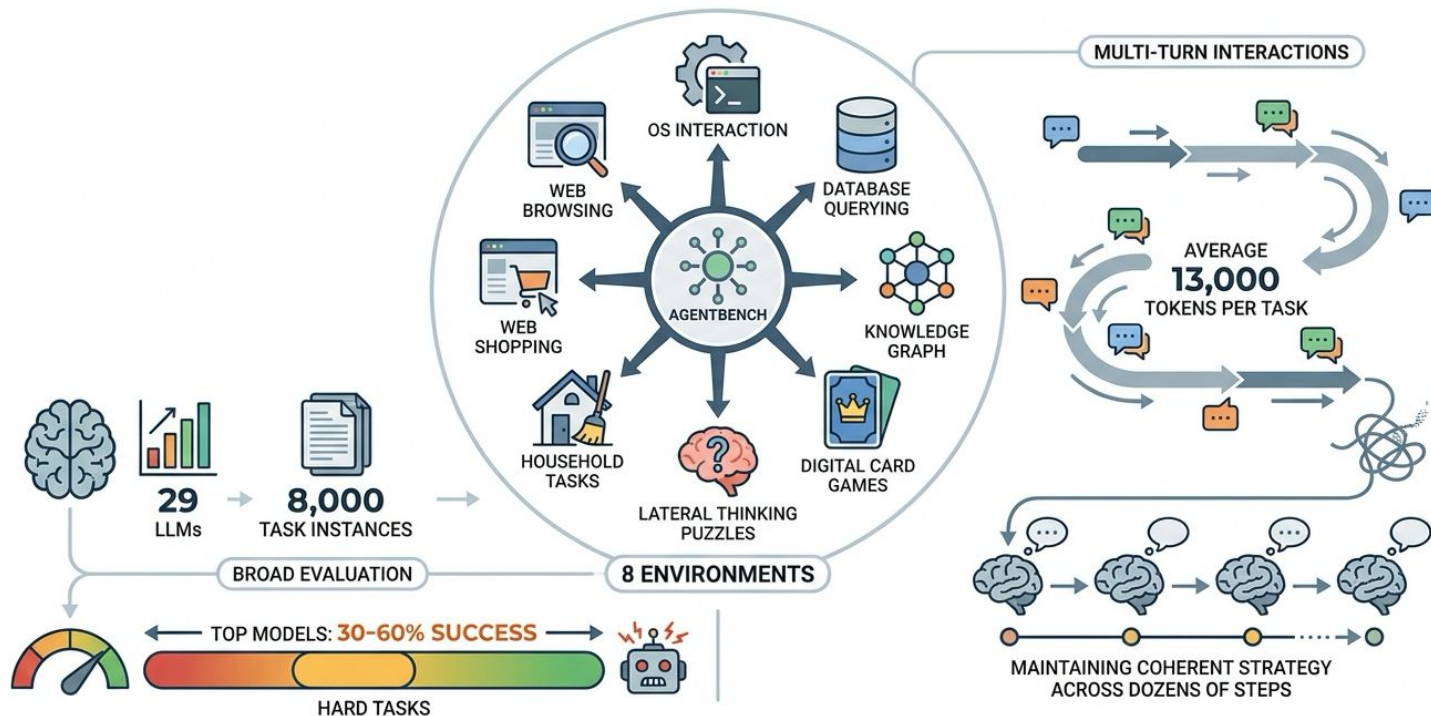
LEADERBOARD GAINS VS. PRODUCTION VALUE



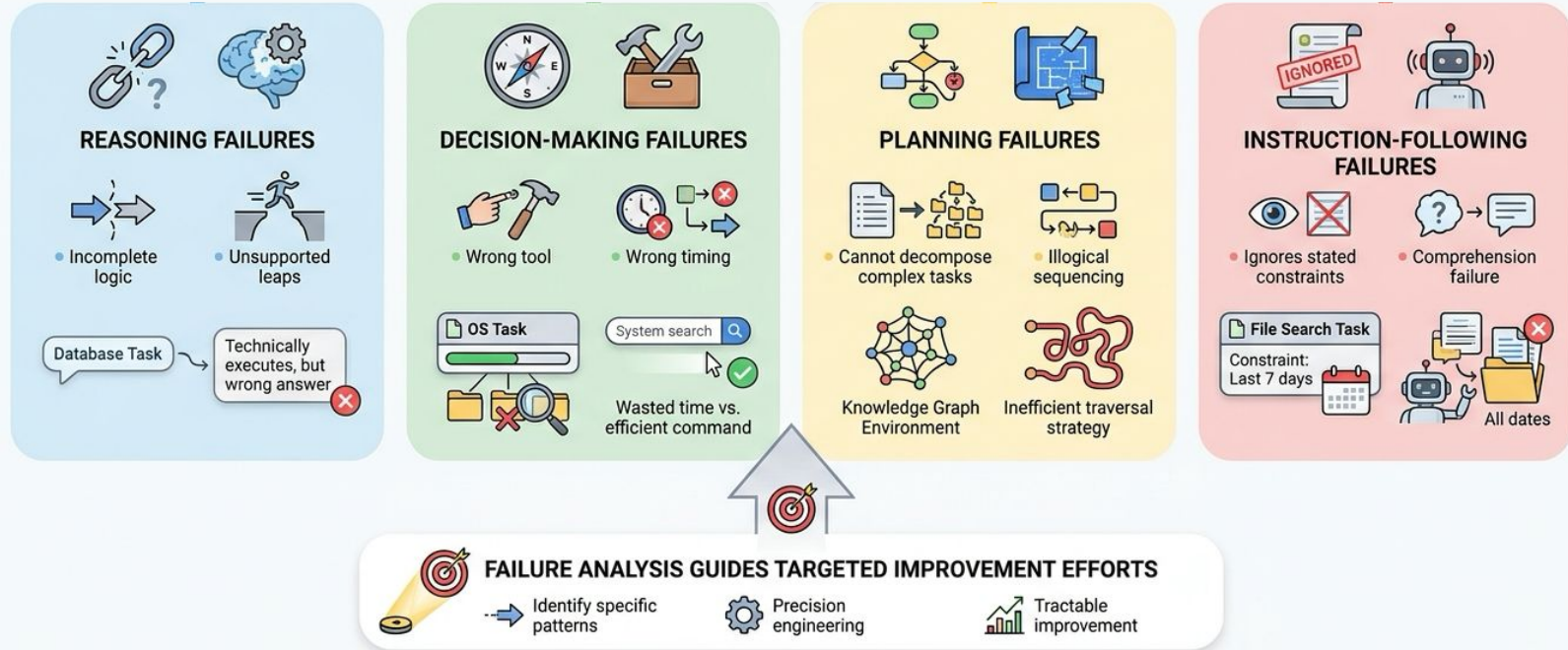
High leaderboard scores often do not transfer to actual production performance.

The solution: Triangulate across multiple evaluation sources, including diverse benchmarks, generalization testing, production pilots, and user studies.

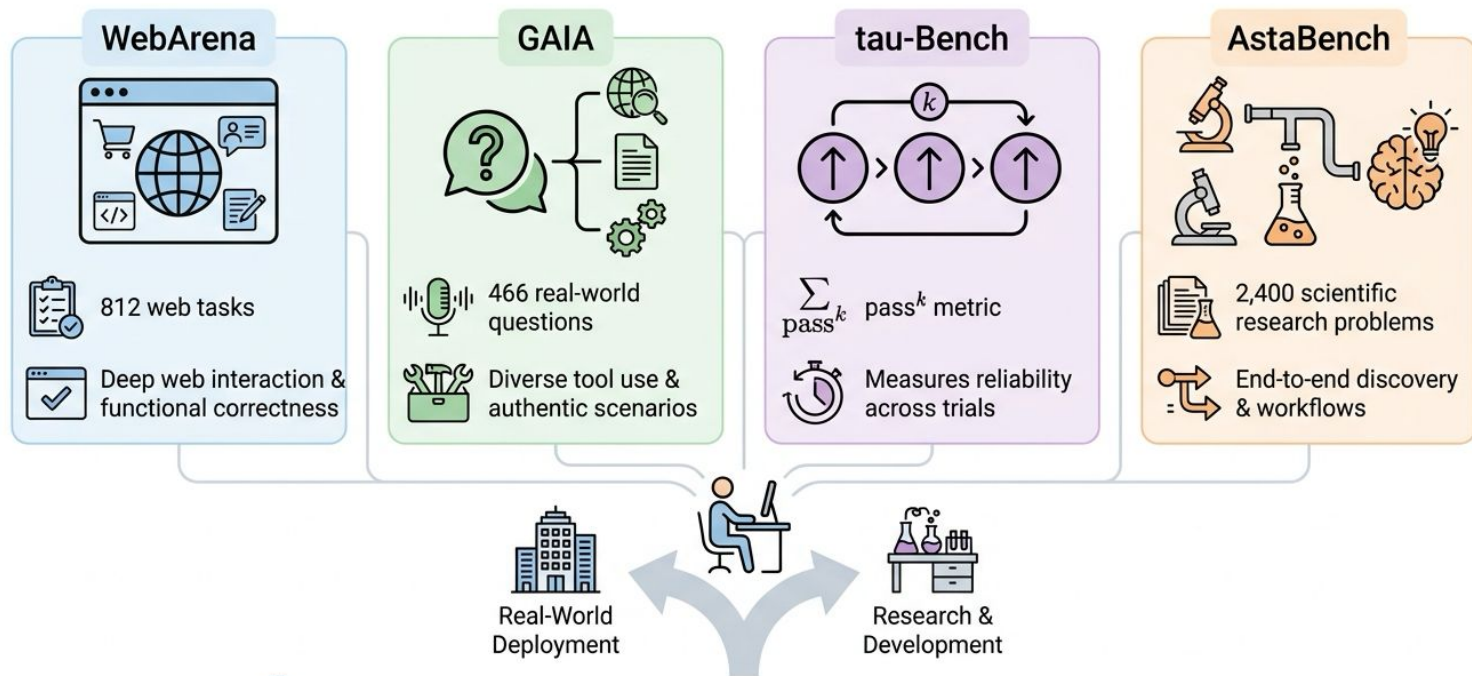
AgentBench -- Eight Environments, One Framework



Four Failure Categories

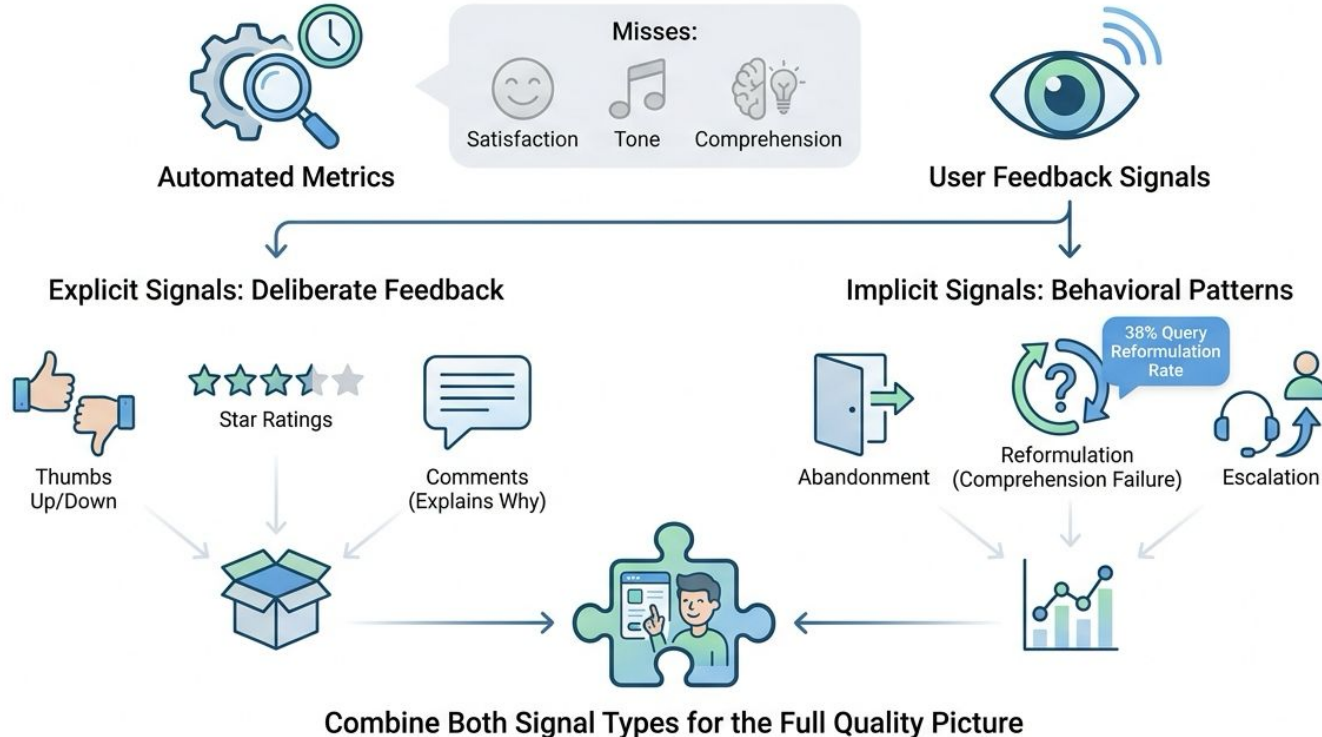


Complementary Benchmark Frameworks

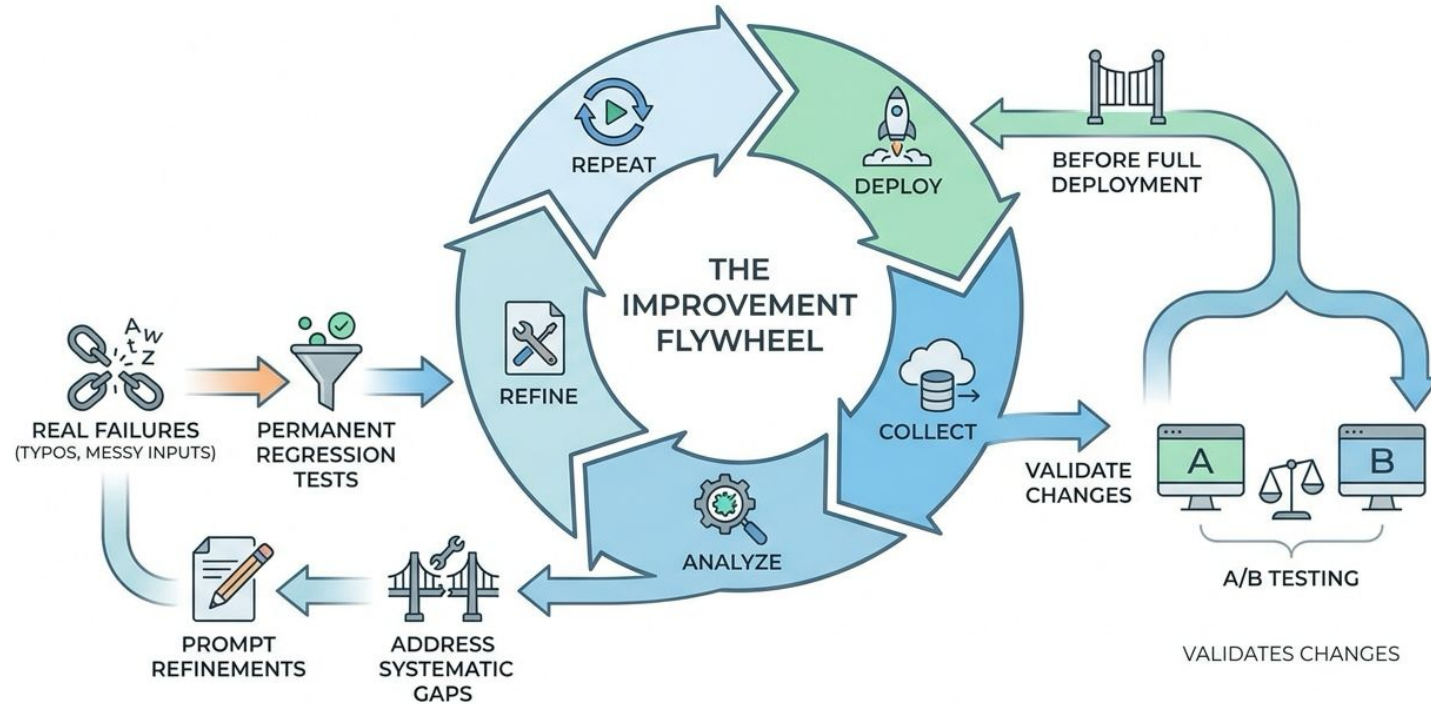


Choose frameworks matching your deployment context.

User Feedback -- What Benchmarks Cannot See



Closing the Feedback Loop



Key Takeaways

- Single benchmarks create a dangerous illusion of readiness
 - Multi-environment frameworks reveal capability distributions
 - Statistical and practical significance both required for decisions
 - User feedback closes the gap between metrics and reality
 - Next: Chapter 3.3 applies these methods to web navigation
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